

Ishraque Zaman Borshon

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SUMMARY

PhD candidate in Aerospace & Mechanical Engineering at the University of Arizona with ~7 years of research and ~3 years of industrial experience. Expertise at the intersection of applied machine learning, computer vision, and scientific/materials engineering: building AI systems for electron microscopy, simulation, and energy-systems data. First-author publications in Q1 journals (npj Computational Materials, Advanced Materials Interfaces, Computational Materials Science) and contributor to production-grade open-source scientific software.

EDUCATION

Ph.D., Aerospace & Mechanical Engineering, University of Arizona Expected July 2026

Tucson, AZ | GPA: 3.5/4.0 | Advisor: Dr. Vitaliy Yurkiv

Dissertation: *Physics-Informed Deep Learning for Atomic-Scale Characterization of Electrochemical Energy Materials*

M.Tech, Energy Systems Engineering, Indian Institute of Technology (IIT) Bombay Jul 2019 – Jul 2021

Mumbai, India | Advisor: Dr. Shireesh B. Kedare | Thesis: *Numerical Study of Cavity Receiver Thermal Losses in Concentrated Solar Power Systems*

B.Tech, Mechanical Engineering, National Institute of Technology (NIT) Kurukshetra Jul 2012 – Jun 2016

Kurukshetra, India | Advisor: Dr. Hari Singh | Senior project: *Design and fabrication of a six-degree-of-freedom industrial robotic hand*

TECHNICAL SKILLS

Machine Learning & DL: Computer vision, Vision Transformers (ViT), U-Net, Mask R-CNN, object detection, semantic segmentation, physics-informed ML, simulation-to-experiment transfer learning, uncertainty quantification, active learning, VAEs, Bayesian optimization, RL/imitation learning, time-series modeling (GRU/LSTM/RNN)

Agentic AI & LLMs: Prompt engineering, LLM fine-tuning, RAG pipelines, vector databases, LangGraph, Claude Agent SDK, multi-agent orchestration, LLM/agent benchmarking

Frameworks & Libraries: PyTorch, PyTorch Lightning, TensorFlow, Keras, scikit-learn, scikit-image, NumPy, Pandas, OpenCV, RLLib, Stable-Baselines3, Matplotlib, Plotly, Streamlit, SciPy

Programming: Python (primary), C, Julia, VB.NET, VBA, C#, MATLAB, C++ (exposure), SQL, Git/GitHub, Linux, LaTeX, Docker (exposure)

Microscopy & CV Analysis: Atomic-column segmentation, defect detection, interface classification, elemental mapping, Fourier/frequency-domain processing, SEM/STEM/TEM (HAADF-STEM, HR-TEM, cryo-TEM), low-contrast image processing

Materials Modeling & Simulation: DFT-based atomistic simulation, multislice STEM simulation (PyQSTEM), grain-boundary modeling, structure-property relationships, ANSYS Fluent (CFD/FEA), COMSOL

HPC & Cloud: GPU-accelerated training on HPC clusters, SLURM, distributed training (PyTorch Lightning), AWS (Bedrock, SageMaker, S3/Redshift), Azure, GCP (exposure)

Electrochemical Characterization: Cyclic voltammetry, EIS, galvanostatic cycling, potentiostatic methods, disordered rocksalt and layered-oxide cathode analysis

Thermal & Mechanical: Thermodynamics, heat transfer, fluid mechanics, refrigerant cycles, 1D/2D/3D numerical heat transfer, HVAC design, GSHP/geothermal systems, concentrated solar power, EnergyPlus

CAD & Design: AutoCAD, SolidWorks, CATIA

RESEARCH & WORK EXPERIENCE

Doctoral Researcher, University of Arizona Aug 2023 – Present

Tucson, AZ | Advisor: Dr. Vitaliy Yurkiv

- Developed Vision Transformer, U-Net, and Mask R-CNN models for atomic-column segmentation, defect detection, and interface classification in STEM/TEM micrographs of electrochemical materials, including SEI layers.
- Built simulation-to-experiment, physics-informed pipelines using multislice STEM simulations (PyQSTEM) and low-contrast image processing for reliable inference on noisy experimental data.
- Predicted column heights and elemental composition in TEM images of high-entropy oxides, linking imaging features to structure-property relationships via DFT-informed ML workflows.

- Investigated solid-electrolyte interphase (SEI) microstructures in lithium-based batteries using atomic-resolution SEM/STEM/TEM analysis; quantified defects and interfaces relevant to degradation, state-of-health, and cycle life.
- Implemented active-learning pipelines for simulation-to-experiment transfer and explored VAEs for generating atomic-scale training data in low-data regimes; applied Bayesian uncertainty quantification for robust inference.
- Developed agentic AI and GenAI-enabled features for engineering simulation workflows, integrating LLM orchestration, RAG pipelines, and prompt-engineering strategies.
- Implemented scalable PyTorch/PyTorch Lightning training and inference on GPU-accelerated HPC clusters using SLURM for large-scale scientific imaging datasets.

Teaching Assistant, University of Arizona

Aug 2024 – Present

Tucson, AZ

- AME 230 Thermodynamics (Spring 2025 & 2026): supported instruction for 114 and 123 students; covered control-volume analysis, property relations, and thermodynamic cycles.
- AME 432 Heat Transfer (Fall 2025): assisted instruction for 97 students in conduction, convection, and radiation; led review sessions and developed worked examples.
- Taught and mentored over 350 undergraduate students across thermodynamics and heat transfer.

Graduate Research Assistant, Oklahoma State University

Aug 2021 – Jul 2023

Stillwater, OK | Advisor: Prof. Jeffrey Spitler

- Co-developed GLHEPro 5.04 & 5.1, commercial ground-loop heat-exchanger design software used by engineers globally; scaled the tool from 308 to over 34,000 borefield configurations using VB.NET.
- Co-developed GHEDesigner, an open-source Python tool for ground-source heat-pump borehole-field design, with OSU and NREL; published an open library of g-functions for 34,321 configurations (IGSHPA 2022).
- Contributed to an ORNL technical report on a techno-economic modeling tool for geothermal heat-pump applications (ORNL/TM-2023/2933); performed sensitivity analysis and data-driven evaluation of system performance.
- Built a Random Forest Regression model to predict heat loss from a Scheffler-concentrator cavity receiver, achieving <1% prediction error on Nusselt number; deployed it as a web-based prediction tool.
- Designed HVAC systems for a 640 sq ft cabin and studied a transcritical CO₂ refrigeration system, finding that a liquid-suction heat exchanger improves COP by ~8x at standard conditions.

Graduate Research Assistant (M.Tech Thesis), IIT Bombay

Jul 2019 – Jul 2021

Mumbai, India | Advisor: Dr. Shireesh B. Kedare

- Designed a high-temperature upward-facing helical cavity receiver for concentrated solar power; performed 3D CFD simulations in ANSYS Fluent to analyze convective and radiative heat loss.
- Derived Nusselt-number correlations from simulation data and benchmarked against literature; built CFD-informed Random Forest models predicting Nusselt number and heat loss with <1% error.
- Deployed the ML model as a web-based engineering prediction tool; analyzed diesel-solar-PV-battery hybrid systems for rural electrification.

Sales Engineer, Petrochem (Bangladesh) Limited

May 2018 – Jun 2019

Dhaka, Bangladesh

- Designed, installed, and serviced solar irrigation and domestic solar PV systems; led business development for 6 foreign companies in the Bangladesh market.
- Led a team of 12 engineers and technicians, contributing to a 5% revenue increase for the company.

Undergraduate Research Assistant, NIT Kurukshetra

Jul 2015 – Jun 2016

Kurukshetra, India | Advisor: Dr. Hari Singh

- Conducted mechanical-engineering research; senior project on the design and fabrication of a six-degree-of-freedom industrial robotic hand.

PUBLICATIONS (FIRST-AUTHOR, PEER-REVIEWED)

- Borshon, I. Z., Amiri, A., Mashayek, F., Shahbazian-Yassar, R., Yurkiv, V. (2026). Evolution of grain boundaries in multi-principal alloys: insights into phase stability and structural complexity. *Computational Materials Science*, 268, 114628.
- Borshon, I. Z., Jabbari, V., Kingston, T. A., Shahbazian-Yassar, R., Yurkiv, V. R. (2025). Deep learning analysis of solid-electrolyte interphase microstructures in lithium-ion batteries. *Advanced Materials Interfaces*, e00558.
- Borshon, I. Z., Ragone, M., Phakatkar, A. H., Long, L., Shahbazian-Yassar, R., et al. (2024). Predicting column heights and elemental composition in experimental TEM images of high-entropy oxides using deep learning. *npj Computational Materials*, 10, 275.

- Spitler, J., West, T., Liu, X., Borshon, I. Z. (2022). Open library of g-functions for 34,321 configurations. IGSHPA Technical/Research Conference.

UNDER REVIEW / IN PREPARATION

- Borshon, I. Z., Kumar, S., Kedare, S. B. (2026). Heat loss mechanisms in helical solar cavity receivers: a combined experimental and 3D numerical study. Solar Energy (under review).
- Borshon, I. Z., et al. (2026). Machine learning-enhanced prediction of heat transfer in helical cavity receivers for CSP (under review).
- Borshon, I. Z., et al. (2026). Machine learning-driven structure-property relationships in high-entropy alloys: a comprehensive review. Progress in Materials Science (under review).

SOFTWARE & TECHNICAL REPORTS

- GHEDesigner [SWR-23-33], open-source Python tool for ground-source heat-pump borehole-field design (OSU & NREL, GitHub).
- GLHEPro 5.04 & 5.1, commercial ground-loop heat-exchanger design software (Oklahoma State University).
- Liu, X., Spitler, J., Mitchell, M., Borshon, I. Z., et al. Final report on an advanced techno-economic modeling tool for geothermal heat-pump applications (ORNL/TM-2023/2933).

CONFERENCE PRESENTATIONS

- Transformer-Enabled HAADF-STEM Quantification of SEI Grain Boundaries for Safety-Relevant Multiphysics Modeling of Li-Ion Batteries. 249th ECS Meeting, Seattle, WA (2026).
- Deep Learning-Based Analysis of Microstructures in SEI and Multi-Principal Element Alloys. MRS Fall Meeting, Boston, MA (2025). Best Poster Nominee.
- Deep Learning Analysis of Solid Electrolyte Interface Microstructures in Lithium-Ion Batteries. PRiME 2024 (ECS), Honolulu, HI (2024).
- Open Library of g-Functions for 34,321 Configurations. IGSHPA Technical/Research Conference, Las Vegas, NV (2022).

LEADERSHIP & SERVICE

- President, Bangladesh Students' Association (200+ members), Oklahoma State University (2023–2024).
- Treasurer, ASHRAE Student Chapter, Oklahoma State University (2022–2023).
- Editorial Board Member, The O'Colly Media Group, Oklahoma State University (2021–2023).
- Representative, Graduate & Professional Student Government Association, Oklahoma State University (2021–2023).

CERTIFICATIONS

- Fundamentals of Accelerated Data Science (NVIDIA RAPIDS, CUDA, cuML).
- Machine Learning with Python; Python Programming & Data Structures (University of Michigan / Coursera).

HONORS, AWARDS & FELLOWSHIPS

- Best Poster Nominee, MRS Fall Meeting, Boston, MA 2025
- John Brammer Graduate Research Fellowship, Oklahoma State University 2021–2023
- Runner-up, CADRE Student Data Awards (Dell Technologies & Intel) 2022
- DAAD Scholarship (German Academic Exchange Service) 2019–2021
- ICCR Scholarship (Indian Council for Cultural Relations) 2012–2016
- Government of India Scholarship (for B.Tech at NIT) 2012–2016
- General Merit Scholarship, top 0.07% (71st of 100,000+), HSC Examination, Bangladesh 2011
- NSF-funded research: Award No. 2311104, Deep Learning-Enhanced TEM Image Analysis for Characterizing Atomic-Scale Structures in Complex Materials.